

CHARACTERIZATION THEORY OF PARACOMPACT SPACES AND BASIC COVERING PROPERTIES

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Preface

Paracompact spaces were introduced by Dieudonné [1944] as a natural generalization of compact topological spaces. On the other hand, fully normal spaces were defined by Tukey [1940] as a generalization of metric spaces. It is perhaps surprising that these two generalizations coincide, since Stone [1948] proved that a Hausdorff space is paracompact iff it is fully normal. In fact, the paracompact spaces provide one of the most successful common generalizations of compact spaces and metric spaces, and thus constitute one of the most important classes of spaces in point-set topology.

Paracompact spaces and the five related classes of covering properties are fundamental. Their characterization theorems are rich and colorful. The study of characterization theory for these basic covering properties has continued for more than 80 years since the publication of Tukey's paper [1940].

This book presents a systematic account of this theory. In Chapters 2 – 4, we present results published between 1940 and 1985, while Chapters 5 – 8 present results obtained by the author. Some of the characterization theorems appeared in Jiang [1987] – [1992] or in the manuscript Jiang [2015]. But the newest results appear here for the first time.

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Contents

Chapter 1 Preliminaries	4
1.1 Basic Covering Properties	4
Index	6

Chapter 1

Preliminaries

1.1 Basic Covering Properties

We use the usual terms and notations^[1]. An ordinal is the set of all smaller ordinals, and a cardinal is an initial ordinal. We use κ, μ for cardinals; λ a for infinite cardinal; $\alpha, \beta, \gamma, \theta$ for ordinal numbers. $\omega = \{0, 1, 2, \dots\}$ is the smallest infinite cardinal. $|A|$ denotes the cardinality of a set A . The empty set ϕ is the unique set B such that $|B| = 0$. $\mathbb{R}$ denotes the real line $[0, 1]$ the closed unit interval in $\mathbb{R}$.

For any sets Z, Y , we denote $\mathcal{P}(Z) = \{A : A \subset Z, \mathbb{Y} = \{f : f \text{ is a function from } \mathbb{Z} \text{ into } \mathbb{Y}\}$. If $f \in \mathbb{Y}$, we write $f : \mathbb{Z} \rightarrow \mathbb{Y}$.

If $f : \mathbb{Z} \rightarrow Y$, $A \subset \Xi$ and $B \subset Y$, the $f(A) : A \rightarrow Y$ defined by $f(A(x)) = f(\infty)$ for each $x \in A$; $f(A) = \{f(x) : x \in A\}$. $f^{-1}B = \mathcal{F}(B) = \{x \in \mathbb{Z} : f(x) \in B\}$.

For a set A and $n \rightarrow \infty$, let $[A]^n = \{B \subset A : |B| = n\}$,

$[A]^{<w} = \cup\{[A]^n : n < w\}$.

Let $w^o = \{\phi\}$, for $n = 1, \dots, w^n =$

$= \{f : f; n \text{ to } w\} = \{(k_0, t_1, \dots, t_{n-1}) : \forall i < n \forall i \in w\}$ where

$m_i = f(i)$. If $t < w$, let $\phi \oplus h = (\overline{t_1}), (k_0, \dots, k_{n-1}) \oplus h =$

$(\overline{k_0}, \dots, \overline{k_{n-1}}, \dots)$. Let $w^{<w} = \cup w^n$.

Let $U = \{U_2 : 2 \in \Delta\}$ be any family of subsets of $\mathbb{Z}$. Without loss of generality we may assume the indexing set Δ be any set of ordinals. We call

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Index

cardinal, §1.1, 4

closed unit interval, §1.1, 4

empty set, §1.1, 4

ordinal, §1.1, 4

real line, §1.1, 4